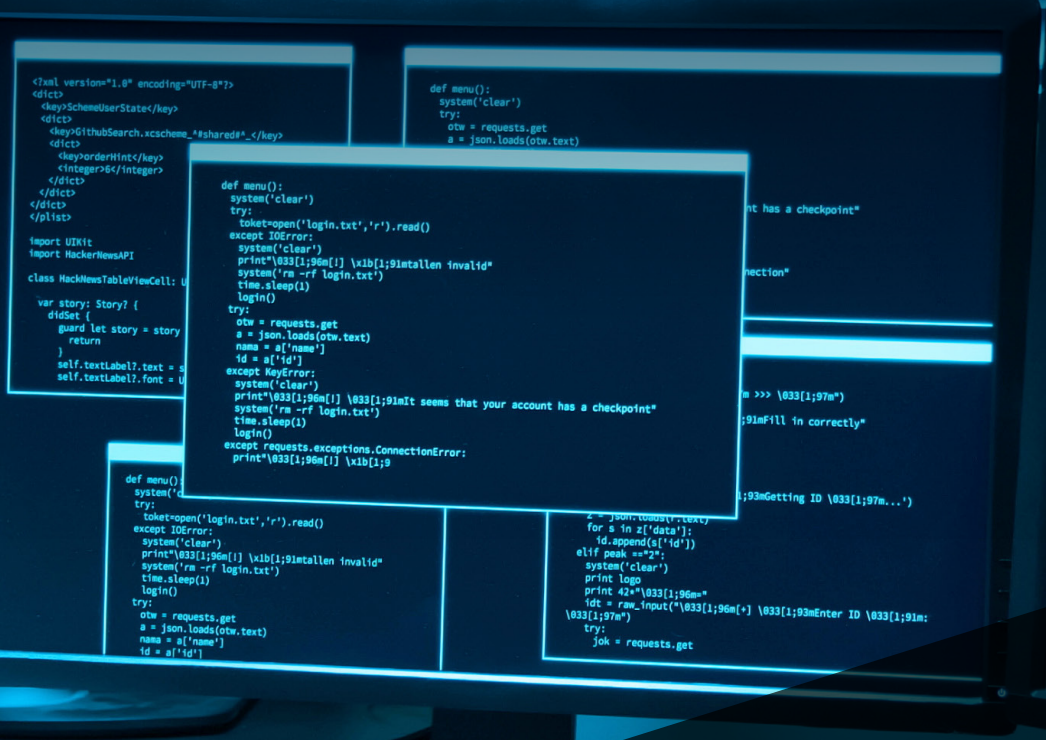


2022

Cybersecurity
INSIDERS

RANSOMWARE REPORT



FORENOVA

INTRODUCTION

Ransomware attacks are on the rise and continue to wreak havoc as one of the most destructive malware threats that affect organizations of all sizes, from SMBs to large enterprises and government agencies, exploiting security vulnerabilities by infecting endpoints and networks through phishing attacks or malicious websites.

Because ransomware is continuously evolving, organizations face significant challenges in responding to the threat and protecting their IT environments against new types of malware. This 2022 Ransomware Report reveals the latest security trends, challenges, and investment priorities for ransomware protection, as reported by over 236 cybersecurity professionals surveyed in 2022.

Key findings include:

- Fifty-three percent of cybersecurity professionals recognize ransomware as their biggest security challenge.
- Ransomware is viewed as a moderate or extreme threat by 88% of survey respondents.
- In the next 12 months, 71% percent of respondents predict ransomware will become an even bigger threat to their organization.
- Confidence in the ability to detect ransomware remains relatively low. A majority of respondents (63%) are at best only moderately confident in their ability to detect and block ransomware before it spreads to critical systems.
- In response to a ransomware attack, most organizations isolate and shut down affected systems and accounts (75%).

We would like to thank [ForeNova](#) for supporting this important research.

We hope you find this report informative and helpful as you continue your efforts to protect your organization against evolving threats.

Thank you,

Holger Schulze



Holger Schulze

CEO and Founder
Cybersecurity Insiders

Cybersecurity
INSIDERS

SECURITY CHALLENGES

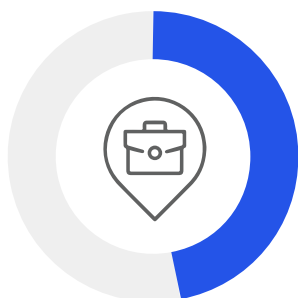
Fifty-three percent of cybersecurity professionals recognize ransomware as their biggest security challenge. The recent shift in work models puts remote workers in second place (47%), followed by limited visibility into cyber threats (41%), rounding out the top three security challenges experienced by cybersecurity professionals.

► What are your biggest security challenges?



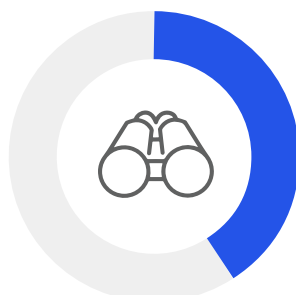
53%

Ransomware



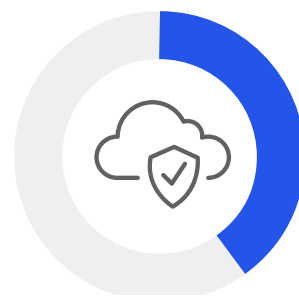
47%

Remote
workers



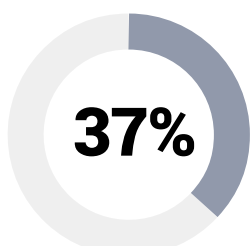
41%

Limited
visibility

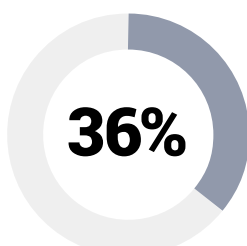


40%

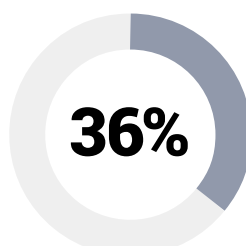
Securing my hybrid
on-prem & cloud
environment



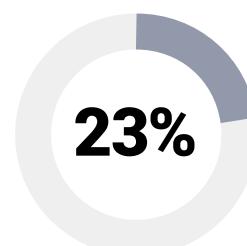
Inconsistent
policy controls



Malware hiding in
encrypted traffic



Too many false
positives/noise



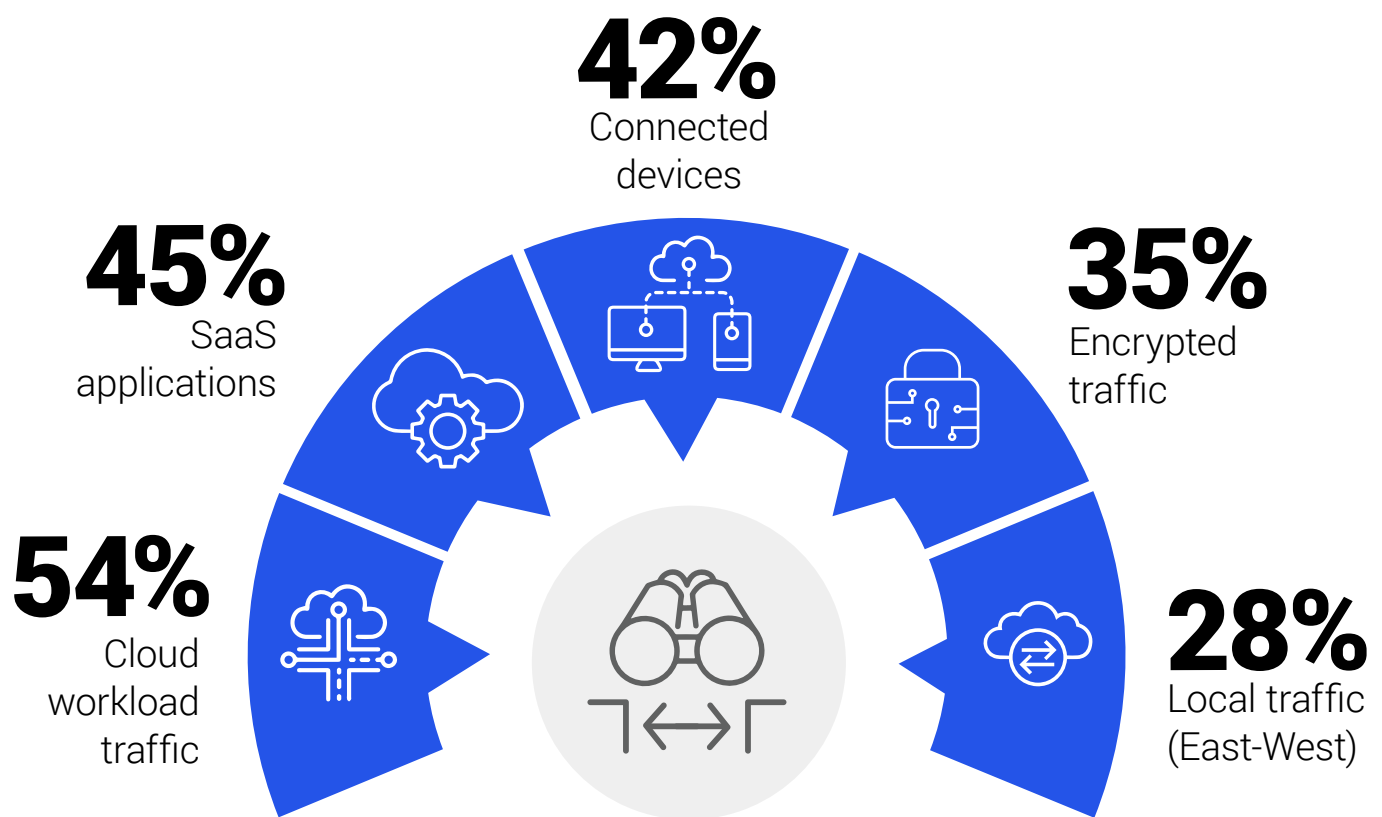
Regaining visibility into
that encrypted traffic

Other 1%

VISIBILITY GAPS

Lack of visibility into networks poses a significant challenge in combating ransomware. Cybersecurity professionals report having the least visibility into cloud workload traffic (54%), followed by SaaS applications (45%), connected devices (42%), and encrypted traffic (35%).

► Where do you have the greatest network visibility gaps?

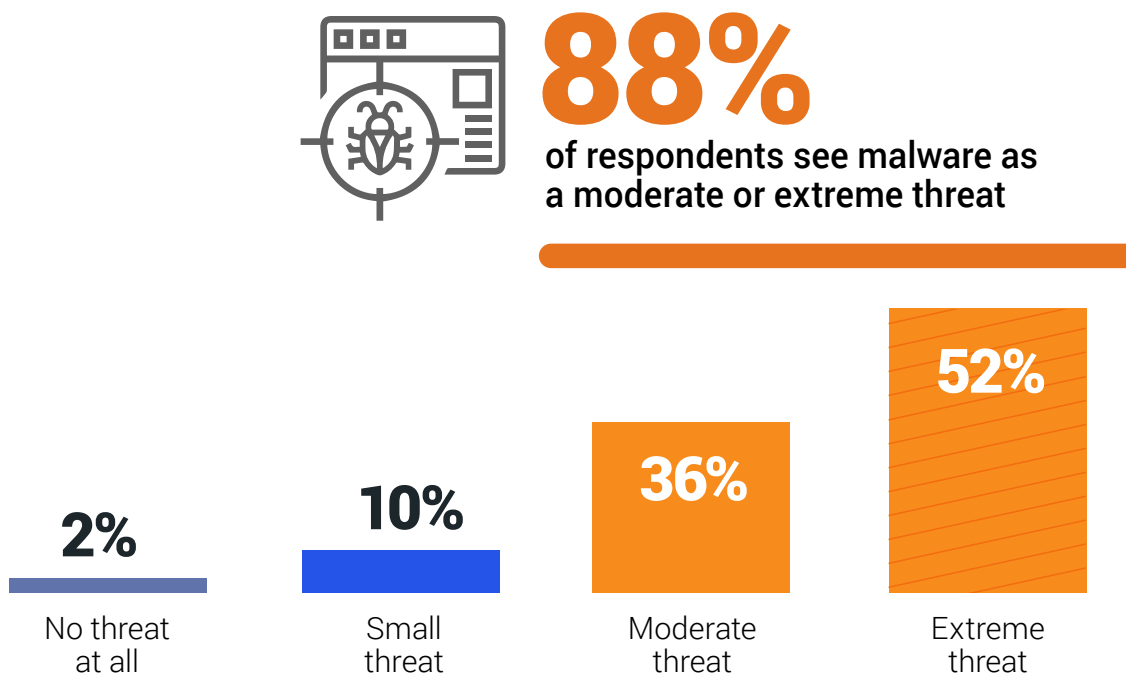


Other 3%

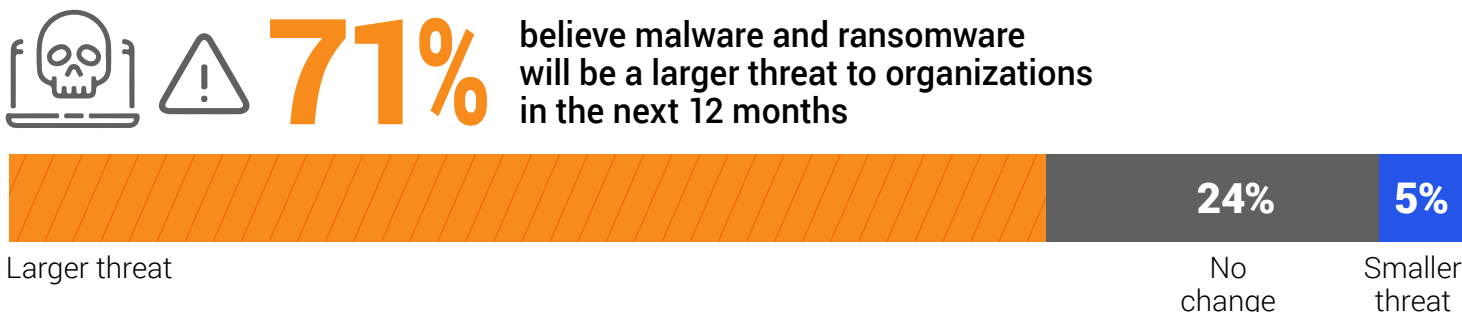
RANSOMWARE THREAT

With ransomware identified as the biggest cybersecurity challenge, we asked how significant a threat it is to businesses. Ransomware is viewed as a moderate or extreme threat by 88% of survey respondents. And the threat is not expected to diminish anytime soon. In the next 12 months, 71% percent of respondents predict ransomware will become an even bigger threat to their organization.

► How significant a threat is malware and ransomware to your business?



► In the next 12 months, do you believe malware and ransomware will be a larger or smaller business threat to organizations?



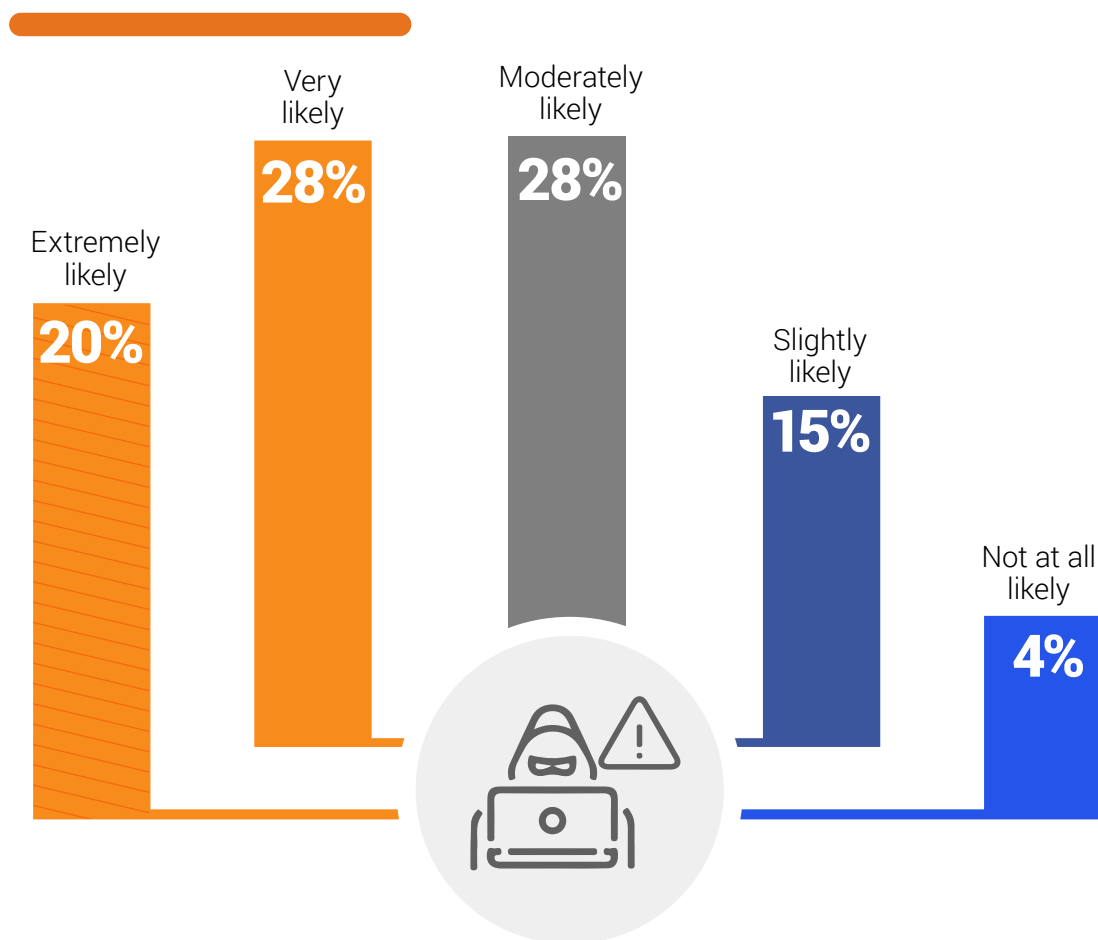
RANSOMWARE THREAT **RISE**

In light of a worsening threat landscape, organizations are quite pragmatic when it comes to ransomware and realize they are being targeted for an attack. Almost half of respondents (48%) believe a ransomware attack is very likely (28%) or extremely likely (20%) to happen in the next 12 months.

- What is the likelihood that your organization will be a target of a malware/ ransomware attack in the next 12 months?

48%

of respondents said an attack
is extremely to very likely

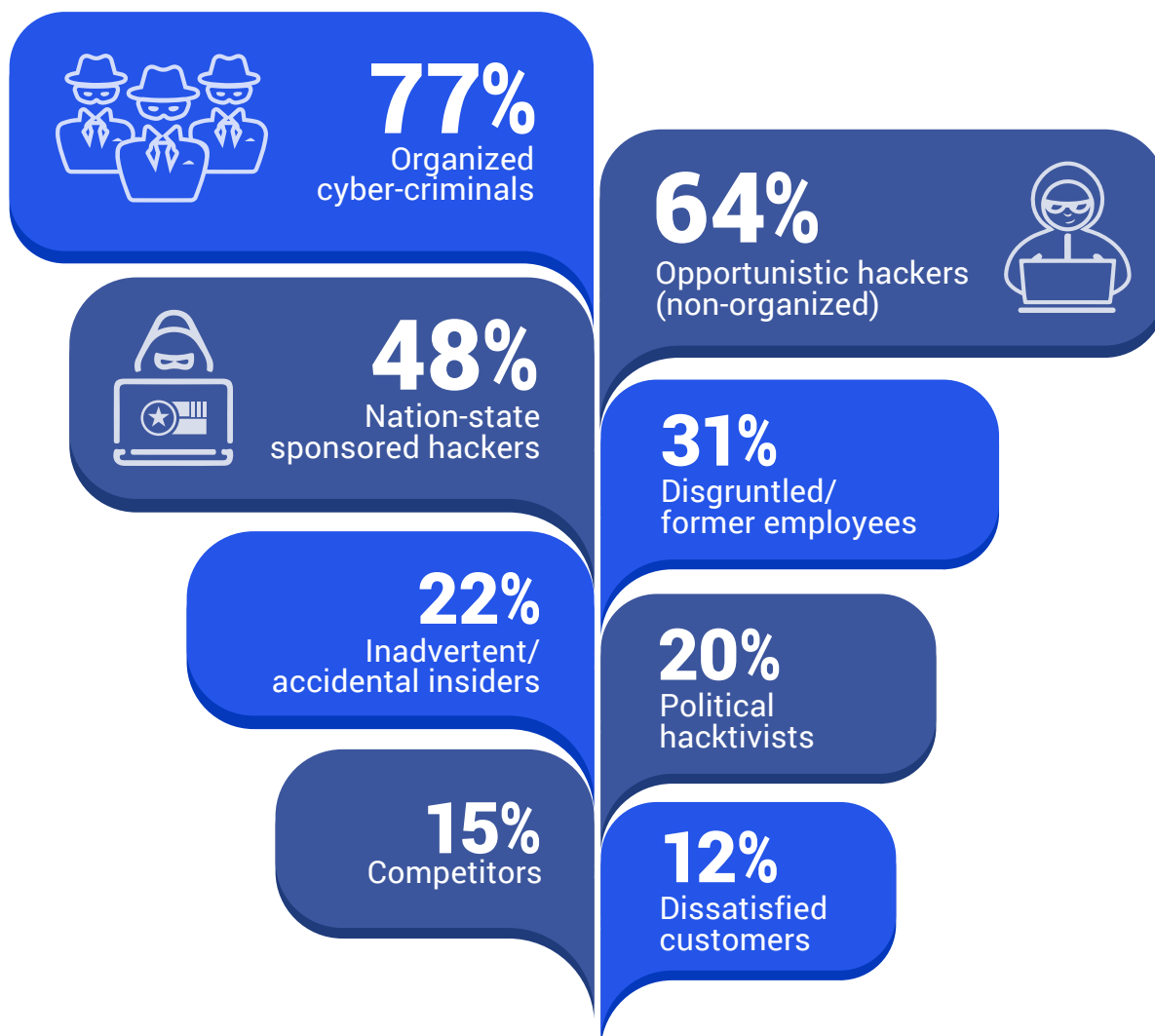


Not sure/other 5%

MOST DANGEROUS ATTACKERS

Cyber attacks can have different sources and motivations. We asked cybersecurity professionals about which attackers concern them the most. Organized cyber-criminals (77%) tops the list, followed by opportunistic hackers (64%), and nation-state sponsored hackers (48%) as the most concerning sources of attacks.

► Which of these attackers concern you most?

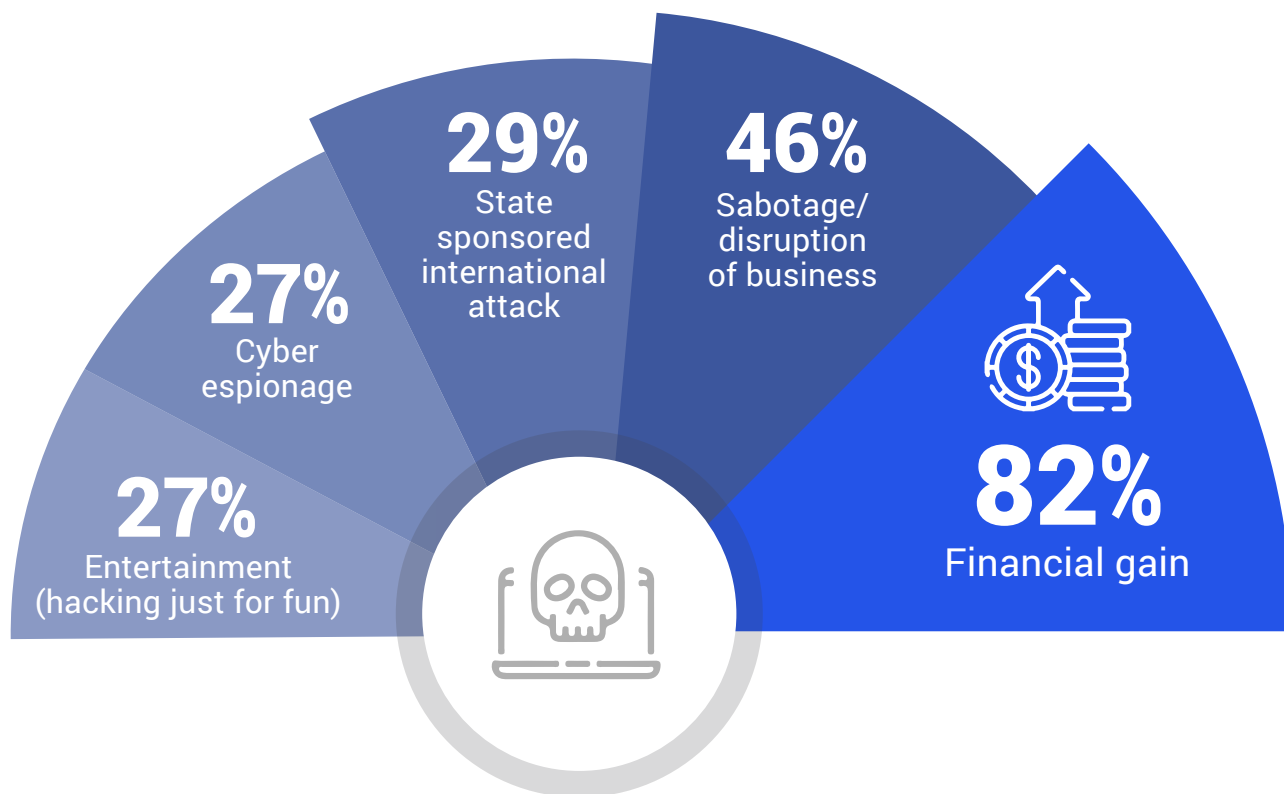


Don't know/other 6%

WHAT MOTIVATES ATTACKERS

So, what do cybersecurity professionals believe motivates attackers to plan and execute ransomware attacks against organizations? Not surprisingly, financial gains tops the list as the main motivator (82%), followed by sabotage and disruption of businesses (46%) and political motives of state sponsors (29%).

- What do you believe are the main motivations for malware/ransomware attacks against your organization?

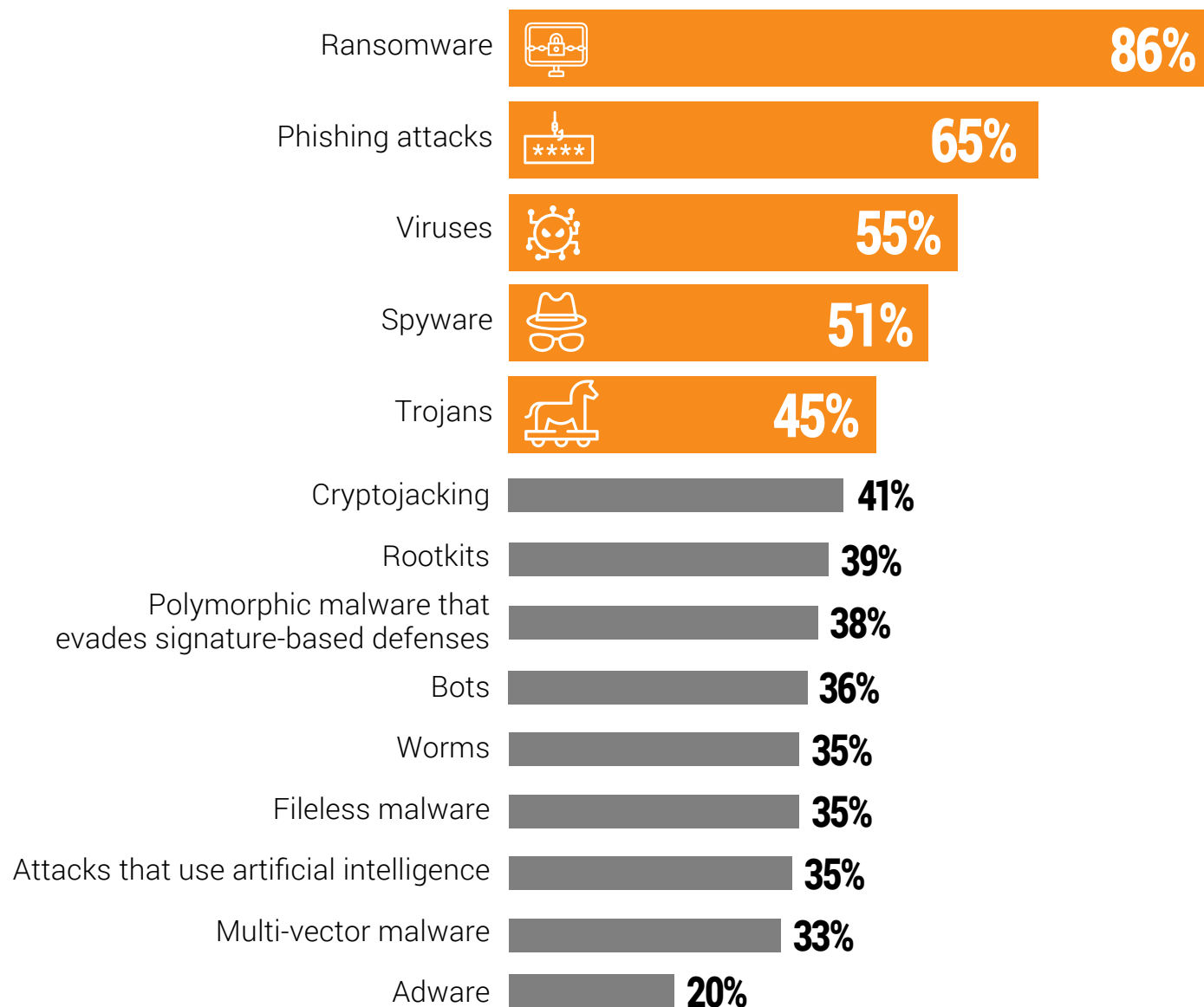


Political motivation 15% | Revenge for a bad experience with organization 10% | Don't know/other 10%

MALWARE TYPES

What types of malware are organizations most concerned about? Not surprisingly, ransomware is the biggest concern (86%), followed by phishing attacks (65%), viruses (55%), and spyware (51%).

► What types of malware are you most concerned about?



Other 1%

MALWARE ATTACK **VECTORS**

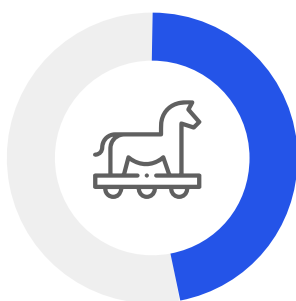
Next, we asked cybersecurity professionals what attack vectors they consider most dangerous. Phishing emails are by far the most mentioned type of attack (79%), followed by trojanized software updates (46%) and web server exploits (42%).

▶ What malware attack vectors do you consider most dangerous?



79%

Spear-phishing
emails



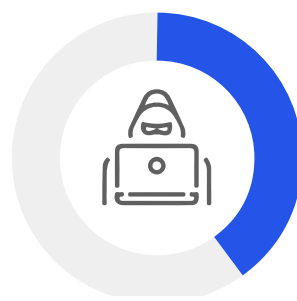
46%

Trojanized
software updates



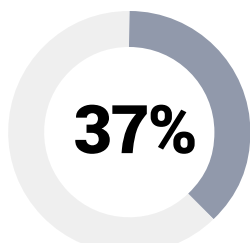
42%

Web server
exploits



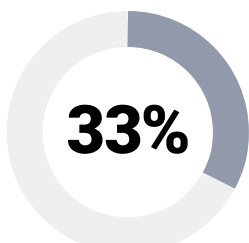
41%

Man-in-the-middle
attacks



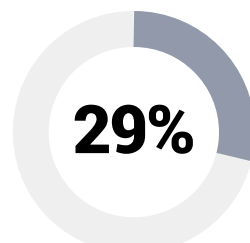
37%

Domain
spoofing



33%

SQL
injection



29%

Cross-Site
scripting

Watering hole websites 25% | "Back door" access to third-party software 24% | Other 2%

HOW RANSOMWARE ENTERS

Ransomware often spreads through phishing emails (58%) that contain malicious attachments or links (52%). This is followed by users unknowingly visiting compromised websites (34%) that load ransomware without the user's knowledge, and ransomware exploiting vulnerable systems (26%).

► How has ransomware entered your organization?

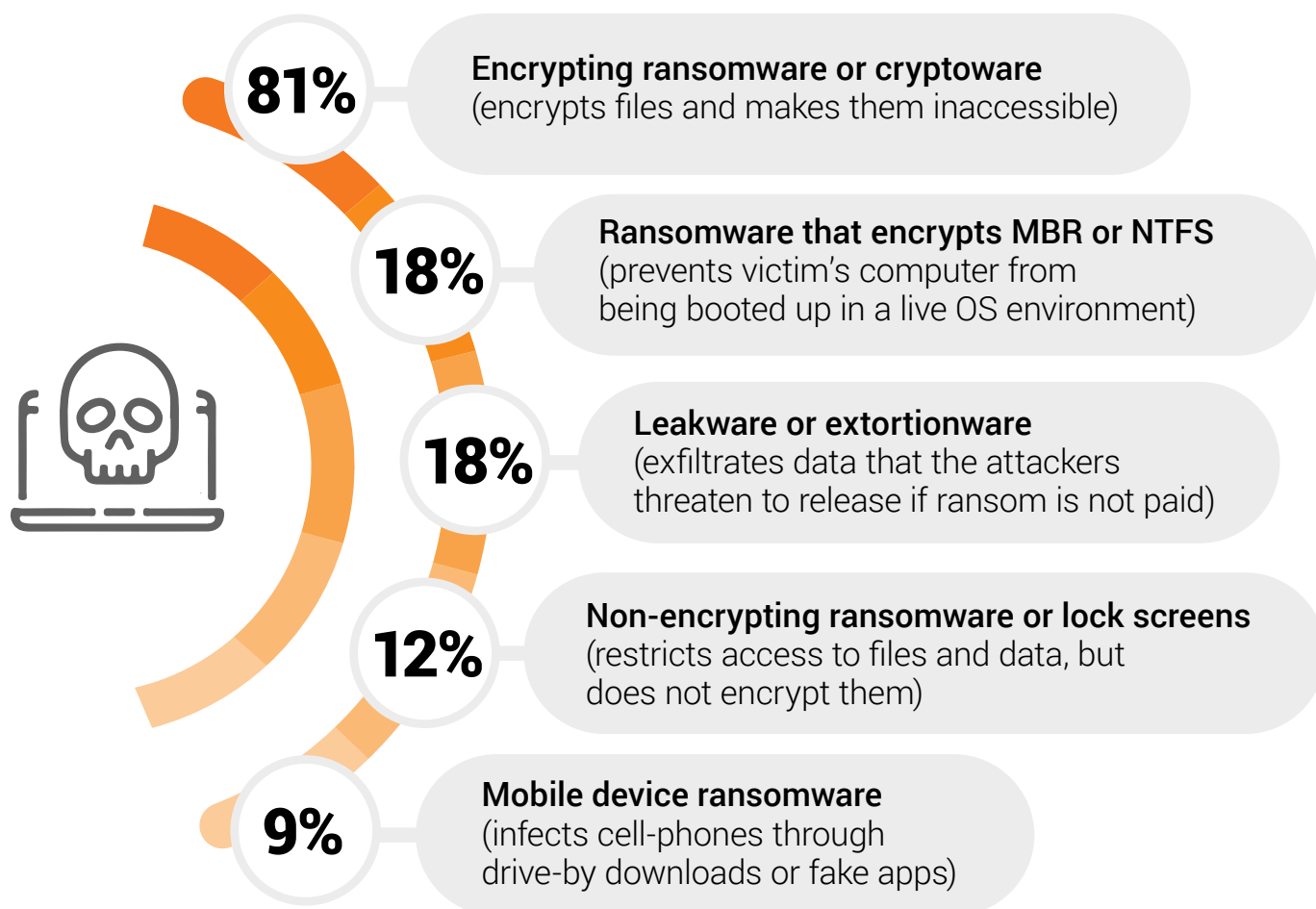


We have never been hit by ransomware 2% | Don't know/other 10%

RANSOMWARE INFECTION

When asked what type of ransomware infected their organization, cyber professionals most frequently report encrypting ransomware or cryptoware as the main culprit (81%).

► What type of ransomware infected your organization?



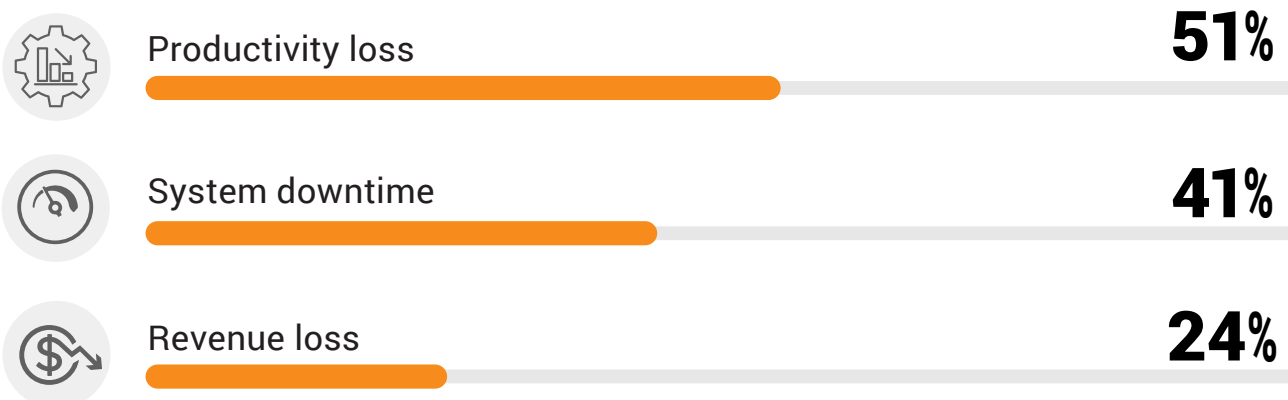
We have not experienced any ransomware infections 3% | Not sure/other 7%

IMPACT OF RANSOMWARE **ATTACKS**

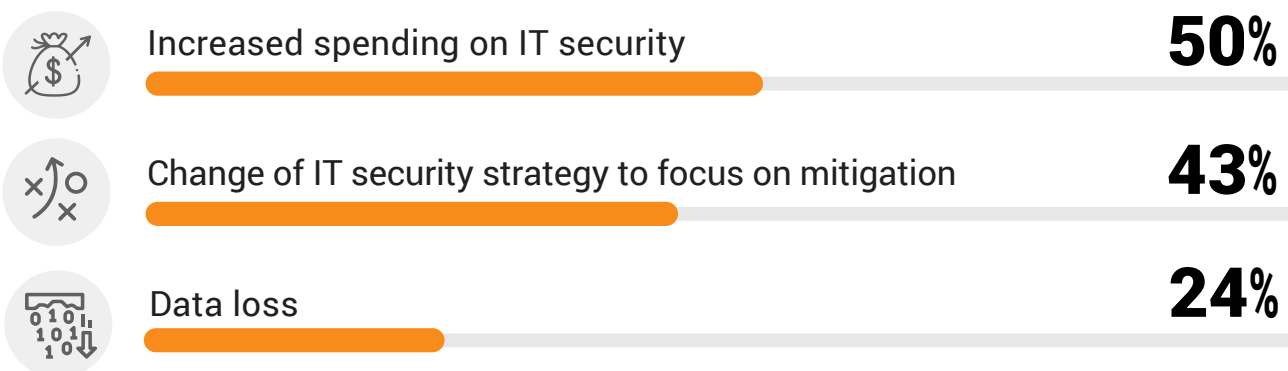
Ransomware attacks directly affect the business. Three of the top five ramifications of a ransomware attack negatively affect the bottom line: productivity loss (51%), system downtime (41%), and revenue loss (24%). The biggest negative impacts on IT operations in response to the ransomware threat include the need for increased spending on security (50%) and changing IT security strategy to focus on mitigation (43%).

► What has been the impact of ransomware attacks on your organization in the past 12 months?

BUSINESS IMPACT



IT OPERATIONS/SECURITY IMPACT

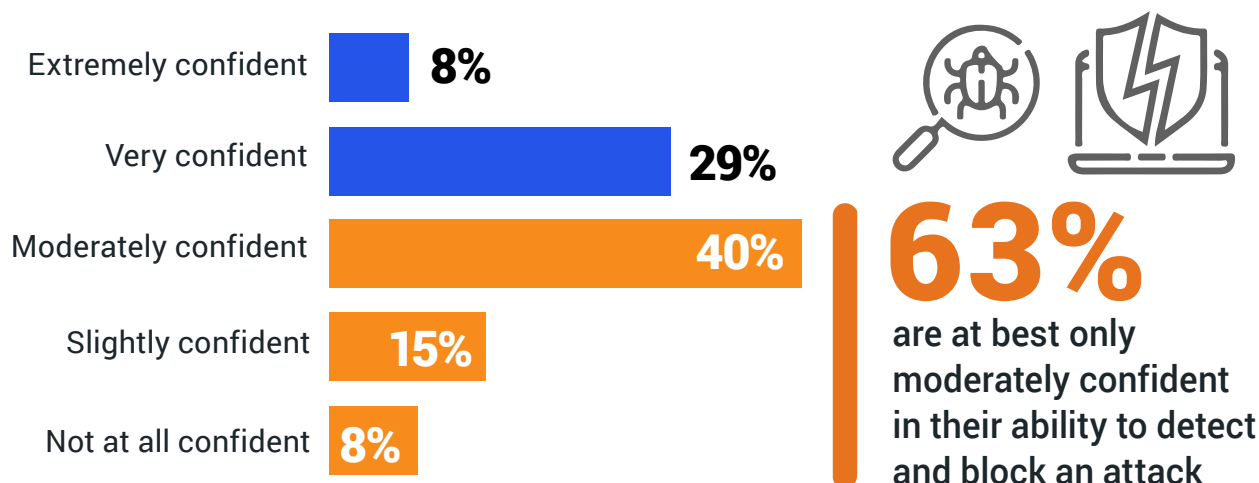


Loss of confidence in existing cybersecurity solutions 21% | We did not experience any ransomware attacks 19% | Negative press/bad publicity 16% | Damage to company reputation 15% | Senior IT staff (CIO, CISO) lost their jobs 10% | Loss of confidence from customers and/or partners 6% | Other 3%

DETECTING RANSOMWARE

Confidence in the ability to detect ransomware remains relatively low. A majority of respondents (63%) are at best only moderately confident in their ability to detect and block ransomware before it spreads to critical systems. In response to a ransomware attack, most organizations isolate and shut down affected systems and accounts (75%).

▶ How confident are you that your organization's defenses are capable of detecting and blocking malware/ransomware before it spreads and infects critical systems and files?



▶ How would your organization respond after a ransomware attack is detected on your systems?



75%

Isolate and shut down offending systems and accounts, recover encrypted files from back-ups, and mitigate the initial attack vector if possible



55%

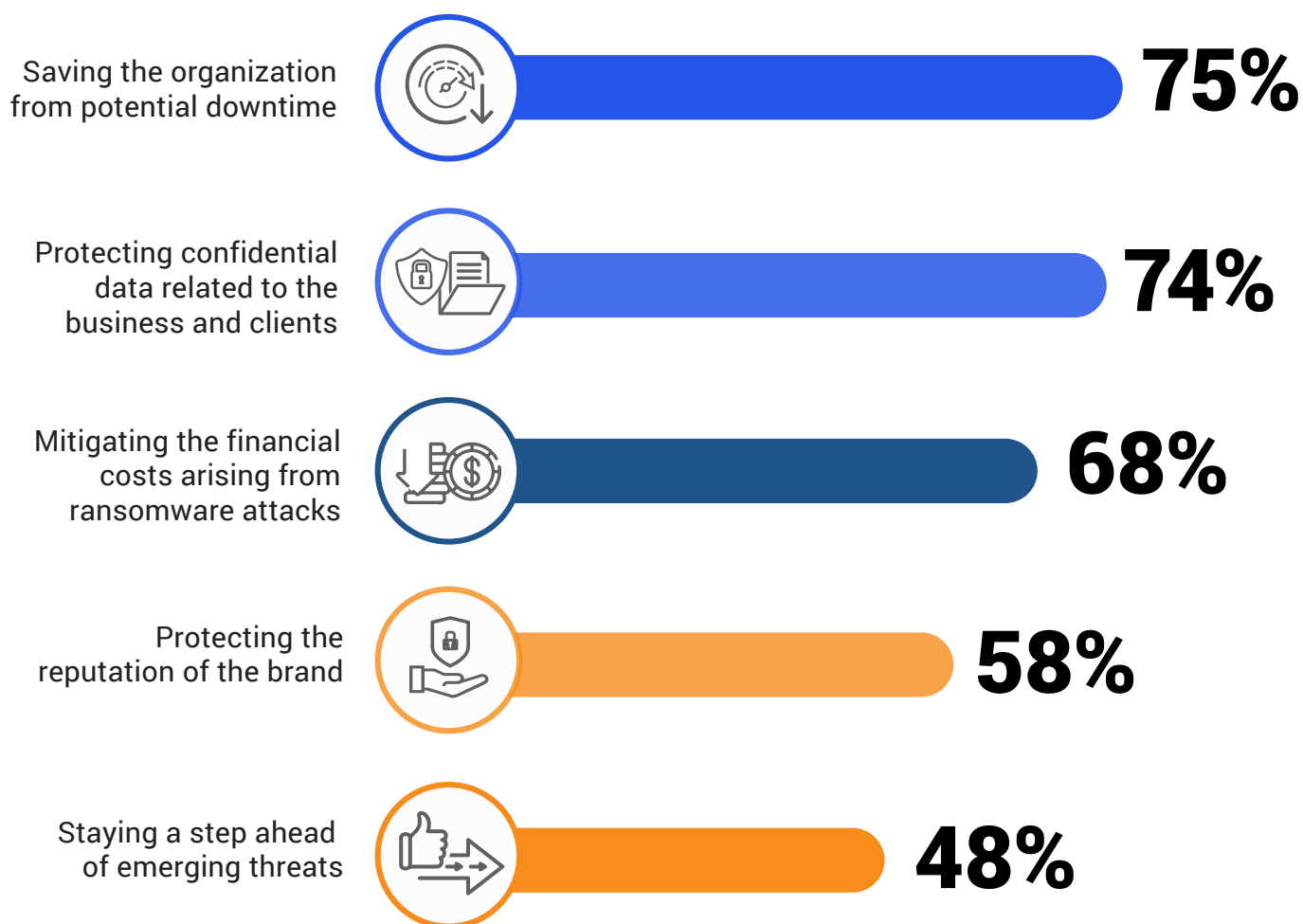
Proactively shut down core systems to prevent spread

Engage a third-party incident response service 39% | Contact cyber insurance provider 38% | Immediately call law enforcement 36% | Contact cybersecurity technology vendor 33% | Attempt to decrypt files ourselves 31% | Notify customers 25% | Attempt to negotiate with the attackers 9% | Pay the ransom 5%

DRIVERS FOR IMPROVING DEFENSES

We asked cybersecurity professionals for feedback on their organization's primary driver for improving ransomware defenses. Saving the organization from potential downtime (75%) and protecting confidential data related to the business and clients (74%) are the most mentioned drivers, followed by mitigating the financial costs of a ransomware attack (68%).

▶ What is your organization's primary driver for improving malware/ransomware defense?



Other 3%

OBSTACLES TO DEFENSE

Security teams are facing a number of obstacles, both internally and externally, in their effort to protect IT environments against ransomware threats. Most organizations report lack of budget (50%) as the primary barrier, followed by the increasing sophistication (49%) and growing proliferation (36%) of attacks as the main obstacles to better ransomware protection.

► What do you believe to be your organization's biggest obstacles to improving malware/ransomware defense?

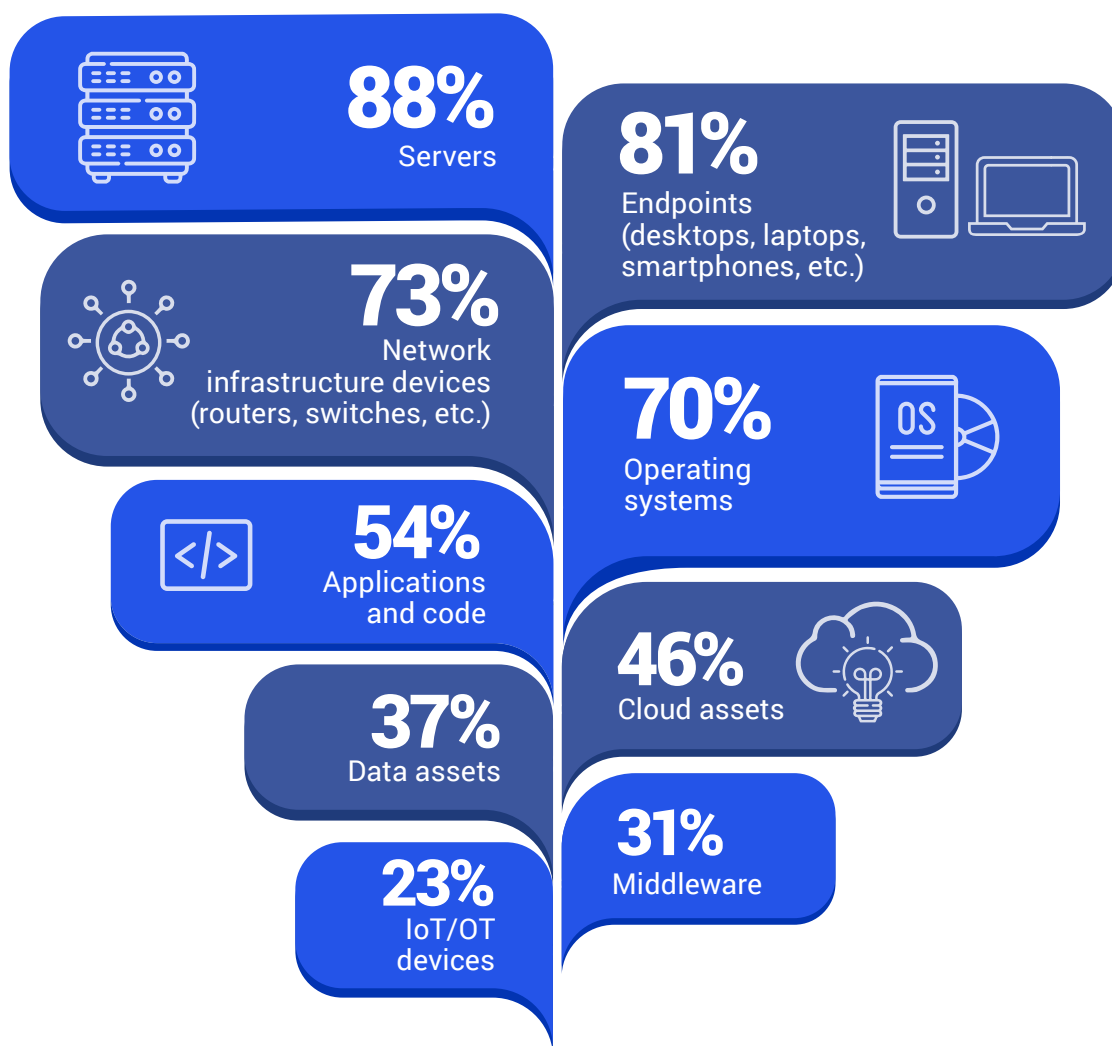


Uncertainty what security solution to use 25% | Lack of executive sponsorship 20% | Our partners' lack of preparedness or response 10% | Other 5%

VULNERABILITY SCANS

Vulnerability scanning of assets across the IT environment allows for quick identification of weaknesses against potential attacks. Organizations most typically scan servers (88%), endpoints (81%), networks (73%), and operating systems (70%) within the IT environment.

► What parts of the IT environment do you typically scan for vulnerabilities?

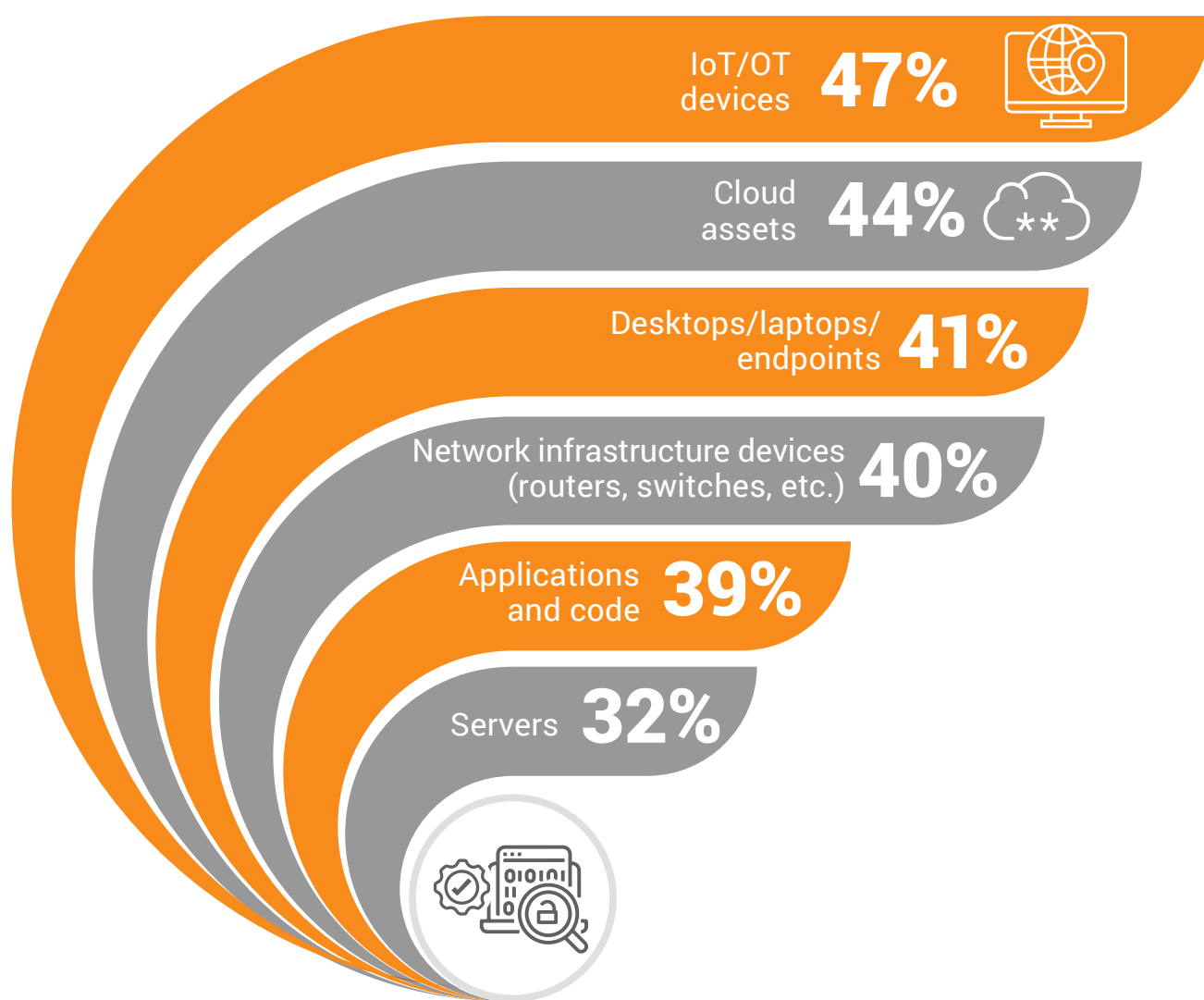


Other 2%

VULNERABILITY MANAGEMENT

We asked cybersecurity professionals which areas of their organization's infrastructure need better vulnerability management. IoT and OT devices tops the list at 47%, reflecting the neglect these devices often experience in most IT environments, from a security perspective. This is followed by cloud assets (44%) and endpoints (41%).

► Which areas of your infrastructure do you think need better/more vulnerability management?

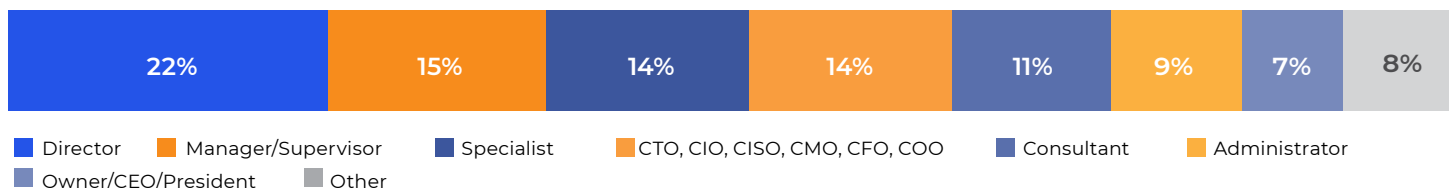


Data assets 31% | Middleware 26% | Operating systems 23% | Other 6%

METHODOLOGY & DEMOGRAPHICS

This report is based on the results of a comprehensive online survey of 236 cybersecurity professionals conducted in Fall 2022, to gain more insight into the latest trends, key challenges and solutions for malware and ransomware security. The respondents range from technical executives to managers and IT security practitioners, representing a balanced cross-section of organizations of varying sizes across multiple industries.

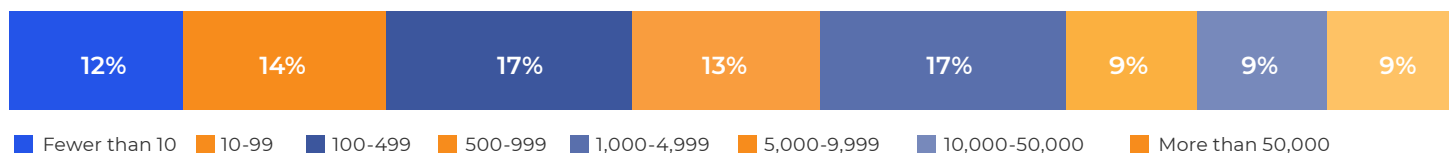
CAREER LEVEL



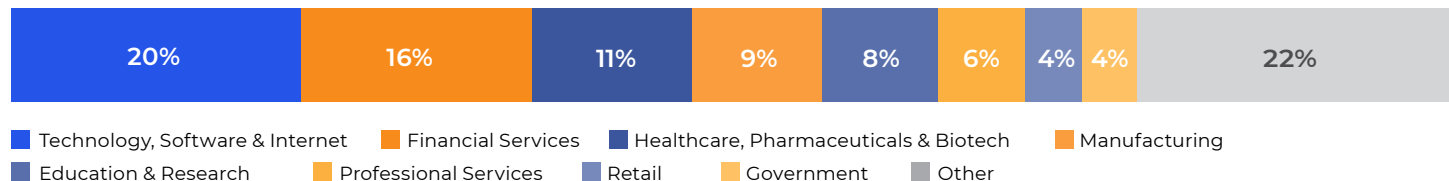
DEPARTMENT



COMPANY SIZE



INDUSTRY





ForeNova: We deliver peace of mind when it comes to Ransomware and other dangerous cyberthreats

We are a rapidly expanding cybersecurity company based in Europe. With our industry-leader-backed security technologies, multi-talented team of security experts, and around the clock cloud security services, we deliver rapid and accurate threat detection and response to help businesses protect the integrity and availability of their critical assets. Our aim is not to replace traditional IT security, but to provide a sophisticated security platform that is prepared to withstand even the most advanced cyber-attacks, today and in the future. Operate your security without concern - with ForeNova.

For more information, please visit www.forenova.com

Cybersecurity

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